US012389214B2

(12) **United States Patent**
**Raghavachari et al.**

(10) **Patent No.: US 12,389,214 B2**
(45) **Date of Patent: Aug. 12, 2025**

(54) **NETWORK DRIVEN PRIVATE AND MACRO NETWORK INTERWORKING**


(71) Applicant: **Verizon Patent and Licensing Inc.**, Basking Ridge, NJ (US)

(72) Inventors: **Balaji L. Raghavachari**, Bridgewater, NJ (US); **Lily Zhu**, Parsippany, NJ (US); **Taussif Khan**, Martinsville, NJ (US)

(73) Assignee: **Verizon Patent and Licensing Inc.**, Basking Ridge, NJ (US)



( * ) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 481 days.

(21) Appl. No.: **17/813,465**

(22) Filed: **Jul. 19, 2022**

(65) **Prior Publication Data**

US 2024/0031791 A1 Jan. 25, 2024

(51) **Int. Cl.**

| | |
|---|---|
| ***H04W 4/00*** | (2018.01) |
| ***H04W 8/08*** | (2009.01) |
| ***H04W 8/18*** | (2009.01) |
| ***H04W 8/24*** | (2009.01) |
| ***H04W 28/02*** | (2009.01) |
| ***H04W 48/10*** | (2009.01) |
| *H04W 84/04* | (2009.01) |

(52) **U.S. Cl.**
CPC ............... ***H04W 8/08*** (2013.01); ***H04W 8/24*** (2013.01); ***H04W 48/10*** (2013.01); *H04W 84/042* (2013.01)

(58) **Field of Classification Search**
CPC ......... H04W 8/08; H04W 8/082; H04W 8/24; H04W 48/10; H04W 84/042; H04W 92/20; H04W 36/0061; H04W 84/105; H04W 88/10; H04W 88/12; H04W 48/20; H04W 88/08; H04W 8/005
See application file for complete search history.


(56) **References Cited**


U.S. PATENT DOCUMENTS

| | | | | |
|---|---|---|---|---|
| 2020/0053821 A1* | 2/2020 | Shih | ...................... | H04W 48/20 |
| 2022/0408352 A1* | 12/2022 | Fujishiro | ............... | H04W 92/20 |
| 2023/0156530 A1* | 5/2023 | Jeong | .............. | H04W 36/00226 370/331 |
| 2023/0284119 A1* | 9/2023 | Kim | ...................... | H04W 60/00 370/329 |

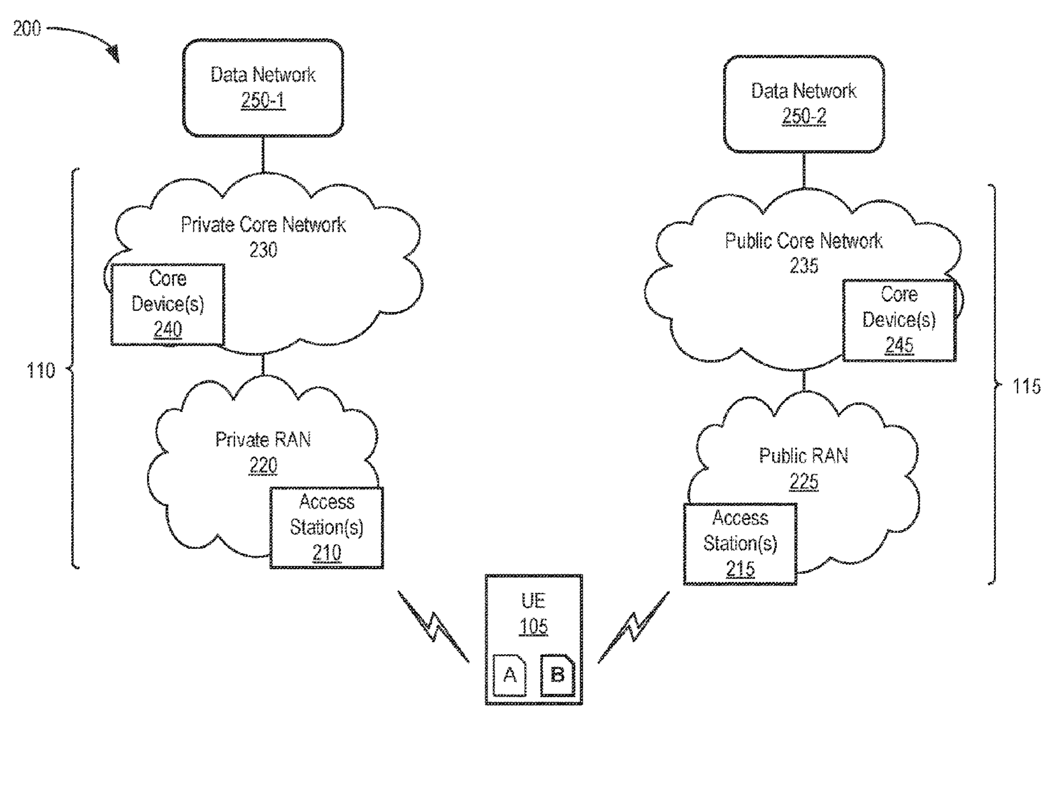
* cited by examiner


*Primary Examiner* — Sai Aung

(57) **ABSTRACT**


Systems and methods described herein provide a network-based solution to efficiently transition private network subscribers from a macro network to a private network. A network device in the macro network stores a neighbor list identifying a private network that has overlapping coverage with a cell for the macro network. The network device detects that a user equipment (UE) device connected to the network device has a subscription to the private network and, in response to the detecting, sends, a release message to the UE device. The release message includes a redirection to the private network.


**20 Claims, 10 Drawing Sheets**



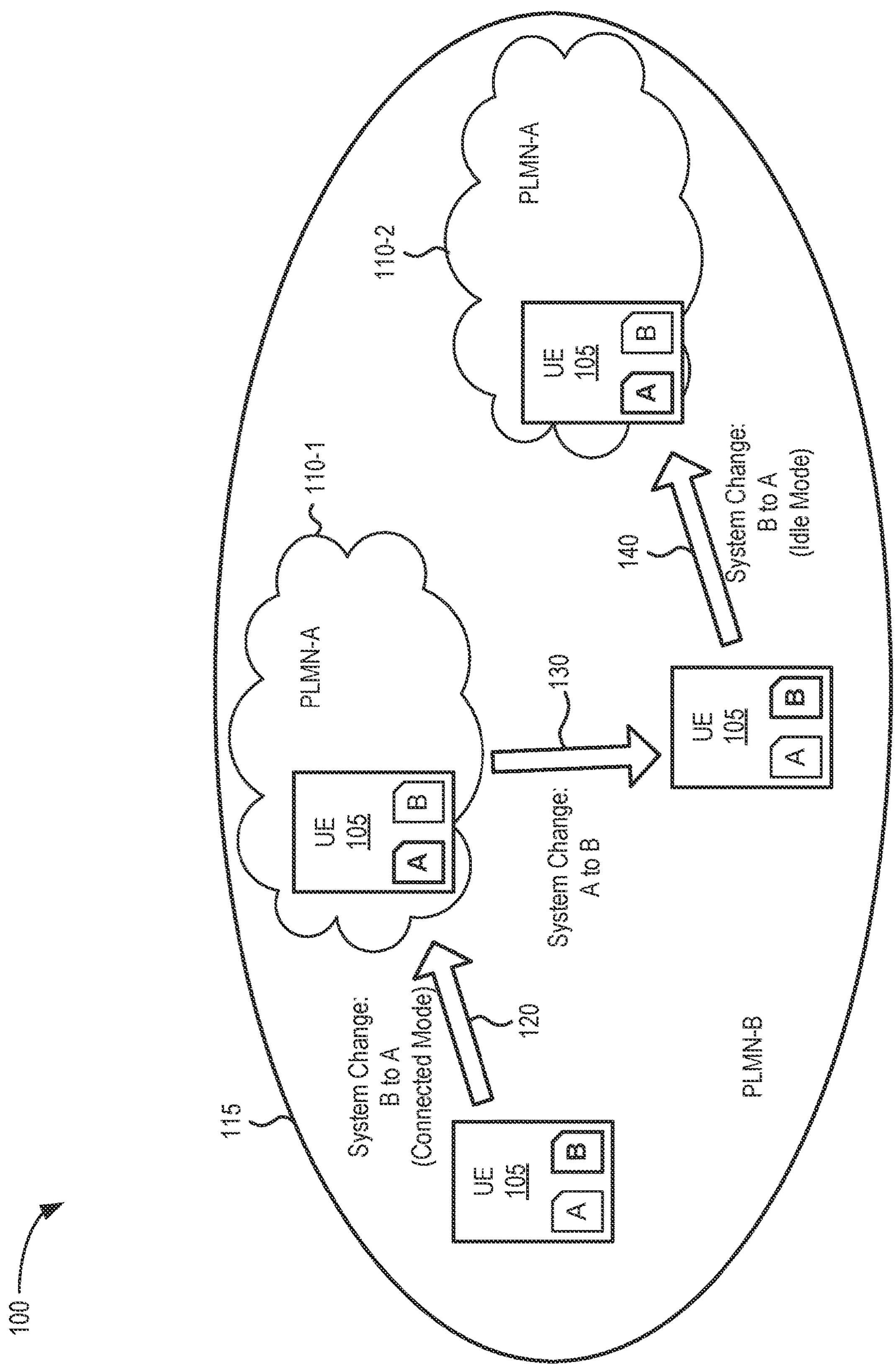
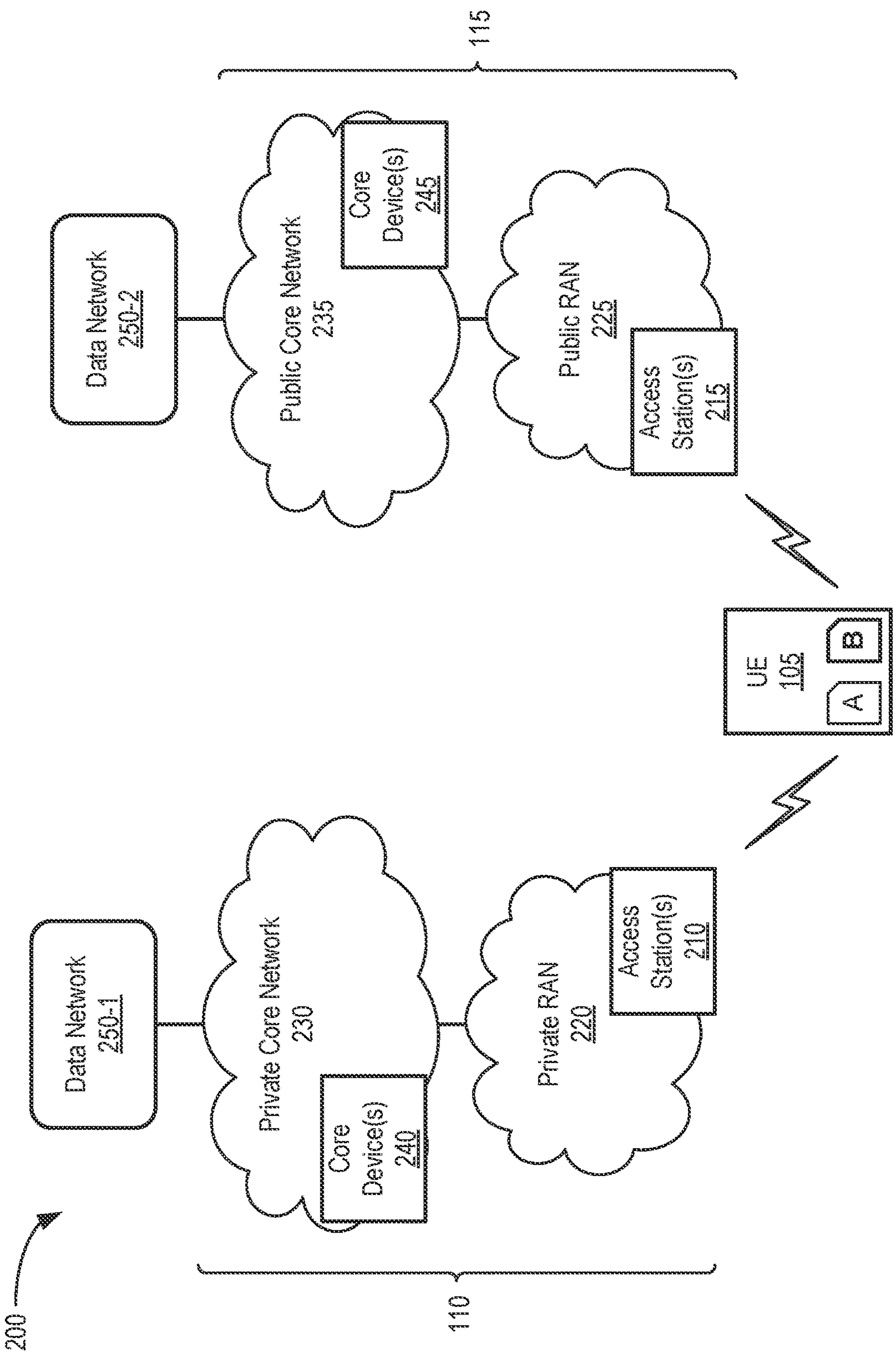

100
115
PLMN-B
UE 105
A
B
System Change: B to A (Connected Mode)
120
PLMN-A
110-1
System Change: A to B
130
140
System Change: B to A (Idle Mode)
PLMN-A
110-2

FIG. 1



200
110
115
Data Network
250-1
Private Core Network
230
Core
Device(s)
240
Private RAN
220
Access
Station(s)
210
UE
105
A
B
Data Network
250-2
Public Core Network
235
Core
Device(s)
245
Public RAN
225
Access
Station(s)
215


FIG. 2

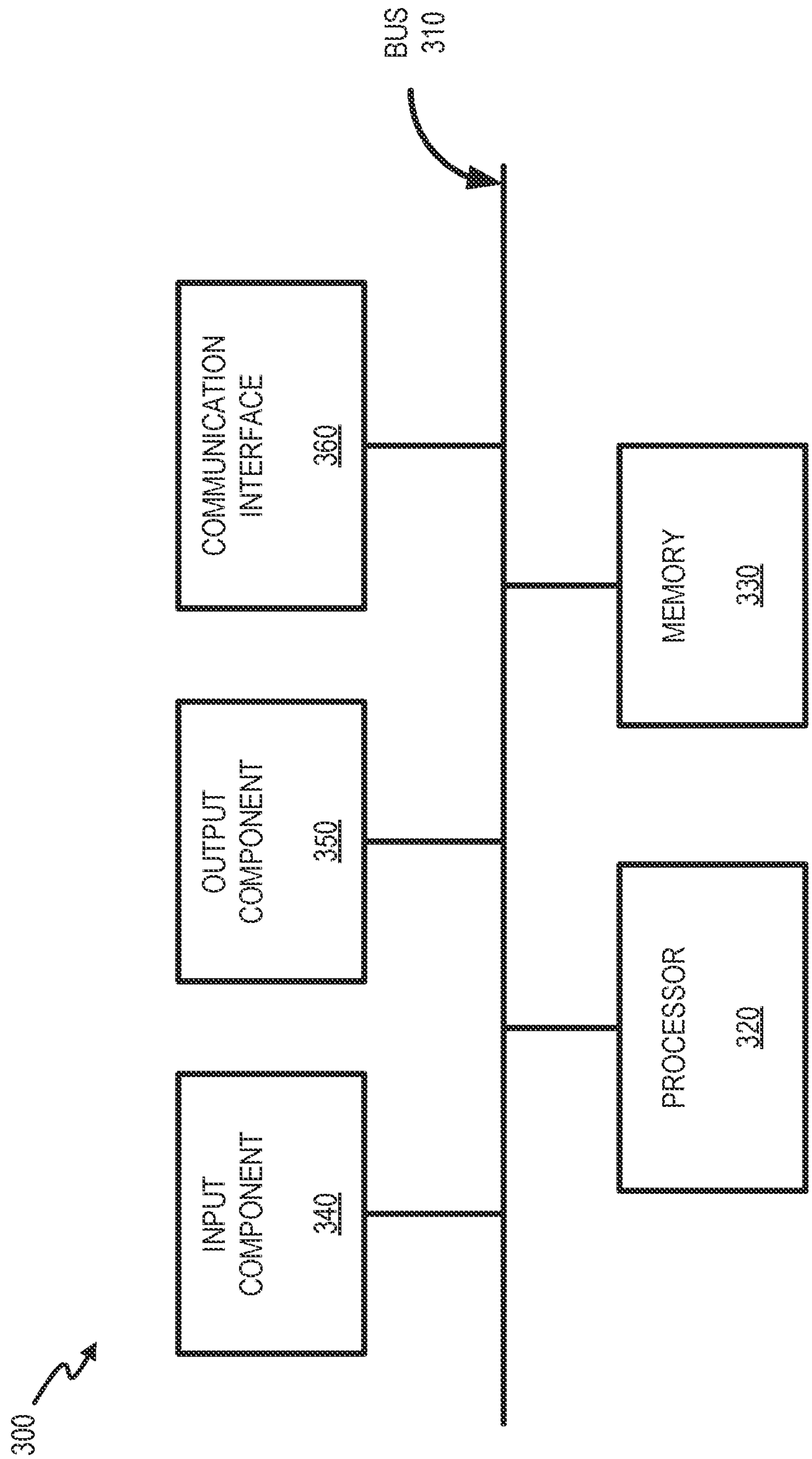

BUS
310
COMMUNICATION
INTERFACE
360
OUTPUT
COMPONENT
350
INPUT
COMPONENT
340
MEMORY
330
PROCESSOR
320
300

FIG. 3

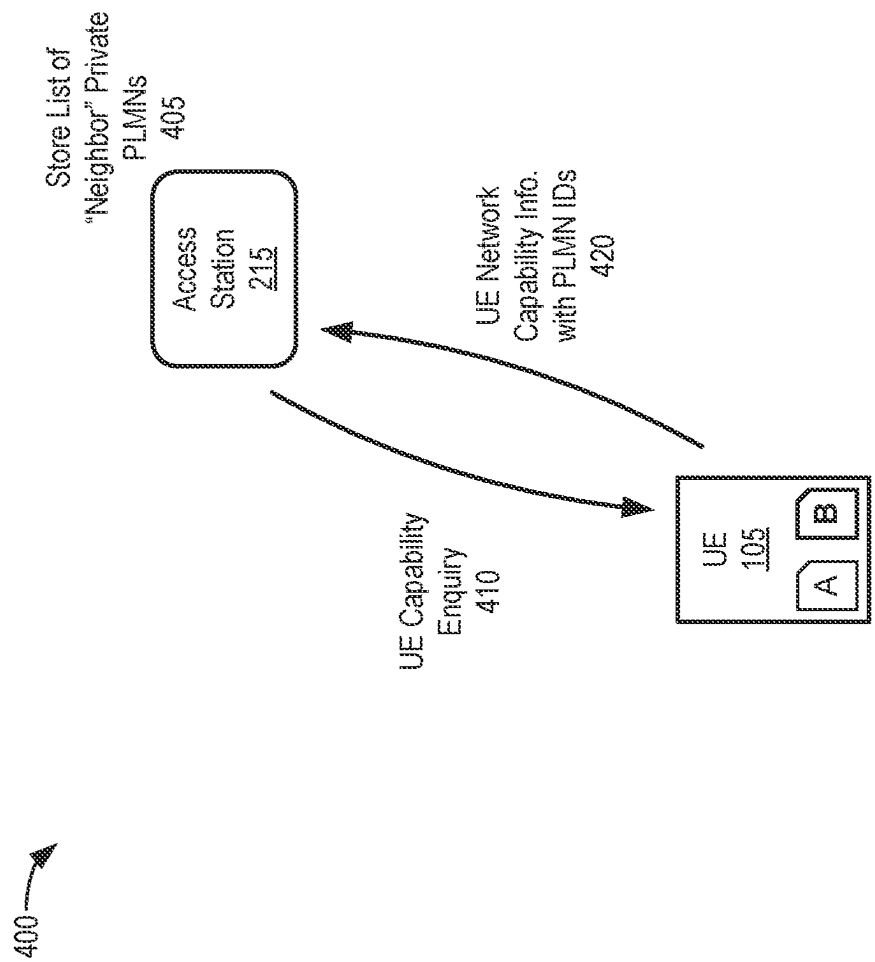

400
Store List of "Neighbor" Private PLMNs 405
Access Station 215
UE Capability Enquiry 410
UE Network Capability Info. with PLMN IDs 420
UE 105
A
B

FIG. 4

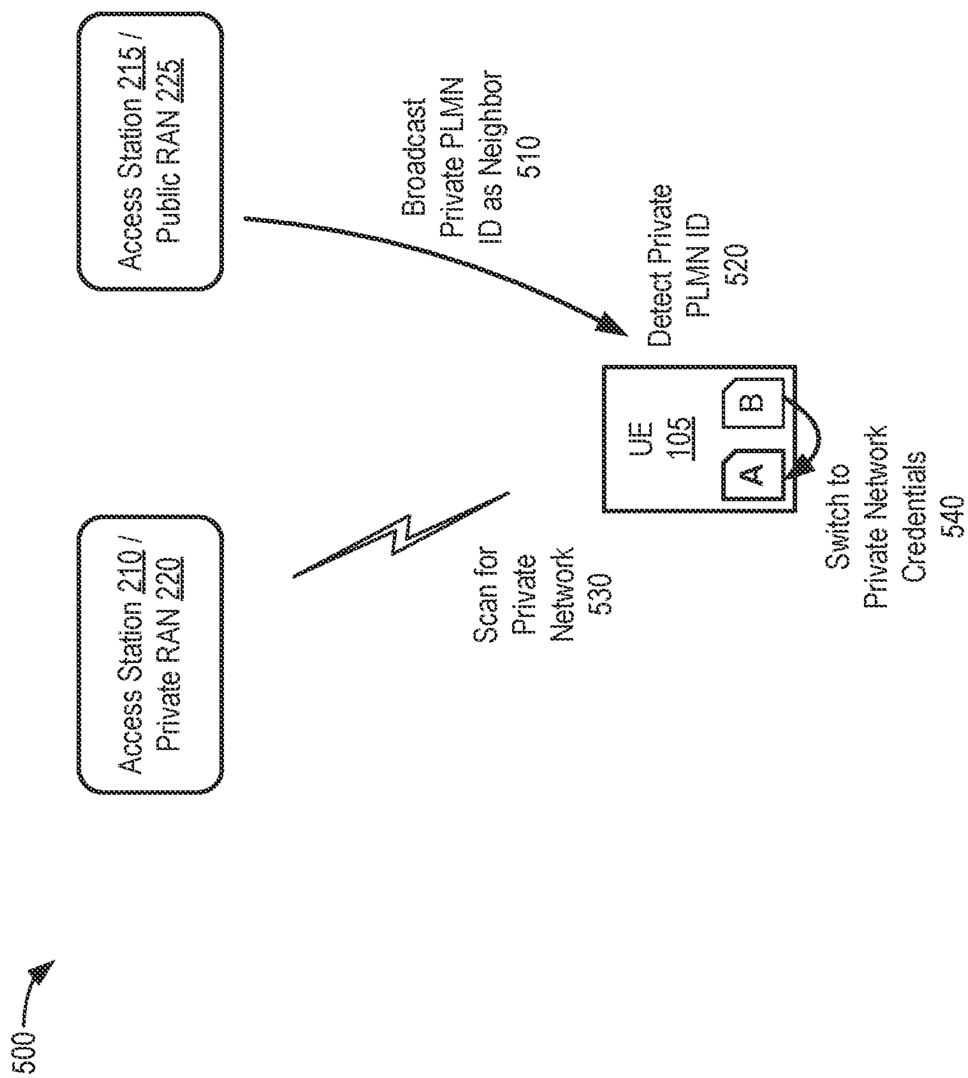
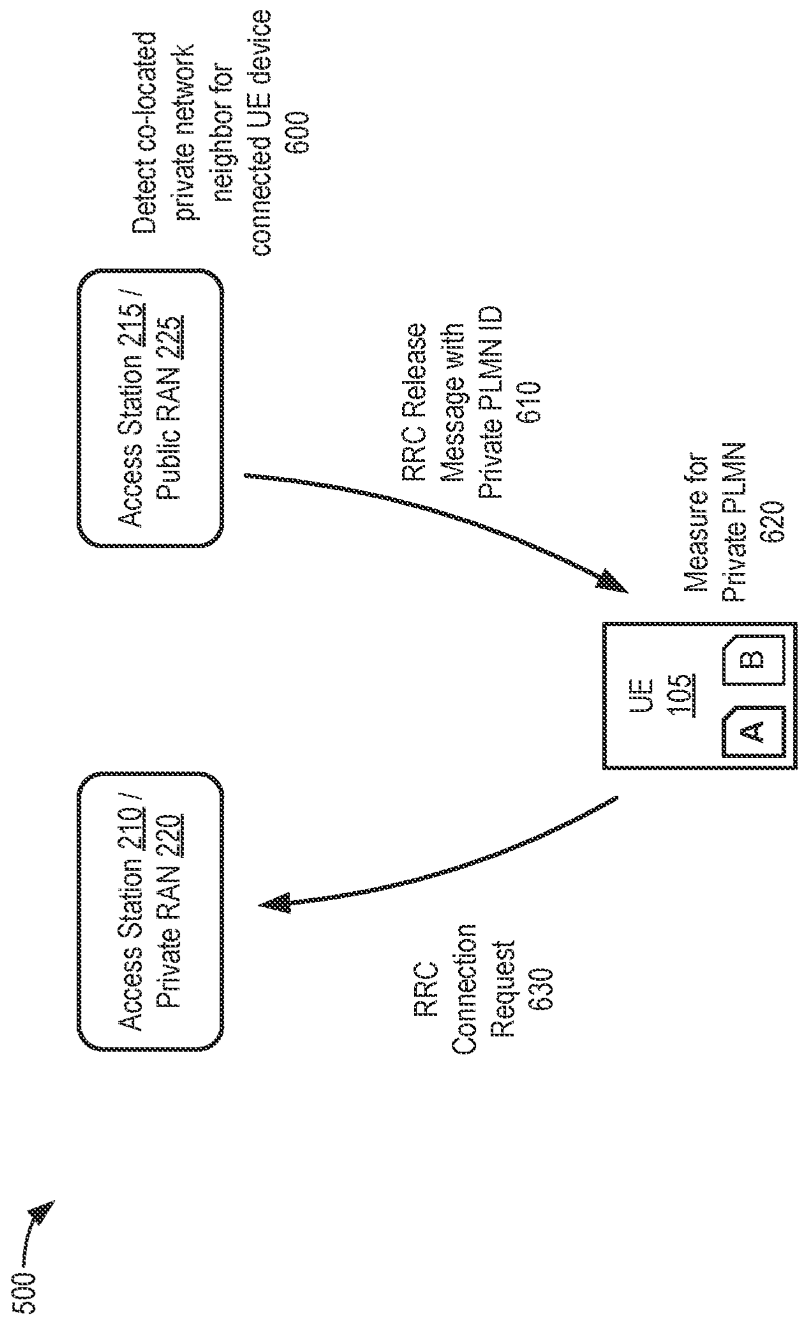

500
Access Station 210 / Private RAN 220
Access Station 215 / Public RAN 225
Broadcast Private PLMN ID as Neighbor 510
Detect Private PLMN ID 520
Scan for Private Network 530
UE 105
A
B
Switch to Private Network Credentials 540

FIG. 5



500
Access Station 215 / Public RAN 225
Detect co-located private network neighbor for connected UE device 600
RRC Release Message with Private PLMN ID 610
UE 105
A
B
Measure for Private PLMN 620
Access Station 210 / Private RAN 220
RRC Connection Request 630


FIG. 6

- SIB25

```
SystemInformationBlockType25-r18 ::= SEQUENCE {
...
networkListNR-r18               NetworkListNR-r18 OPTIONAL, -- Need
t-ReselectionNR-r18               T-Reselection,
t-ReselectionNR-SF-r18            SpeedStateScaleFactors OPTIONAL, -- Need OR
...
}
networkListNR-r18 ::= SEQUENCE (SIZE (1..maxNetwork)) OF NetworkNR-r18
NetworkNR-r18 ::= SEQUENCE {
networkListNR-r18                 PLMN-IdentityList,
.... (rest is the same as carrierFreqList in other neighbor list)
```

710

FIG. 7A

- RRC Connection Release

```
releaseCause ::= ENUMERATED [other]
    redirectedNetworkInfo ::= networkListNR-r18 OPTIONAL, -- Need
```

720

FIG. 7B

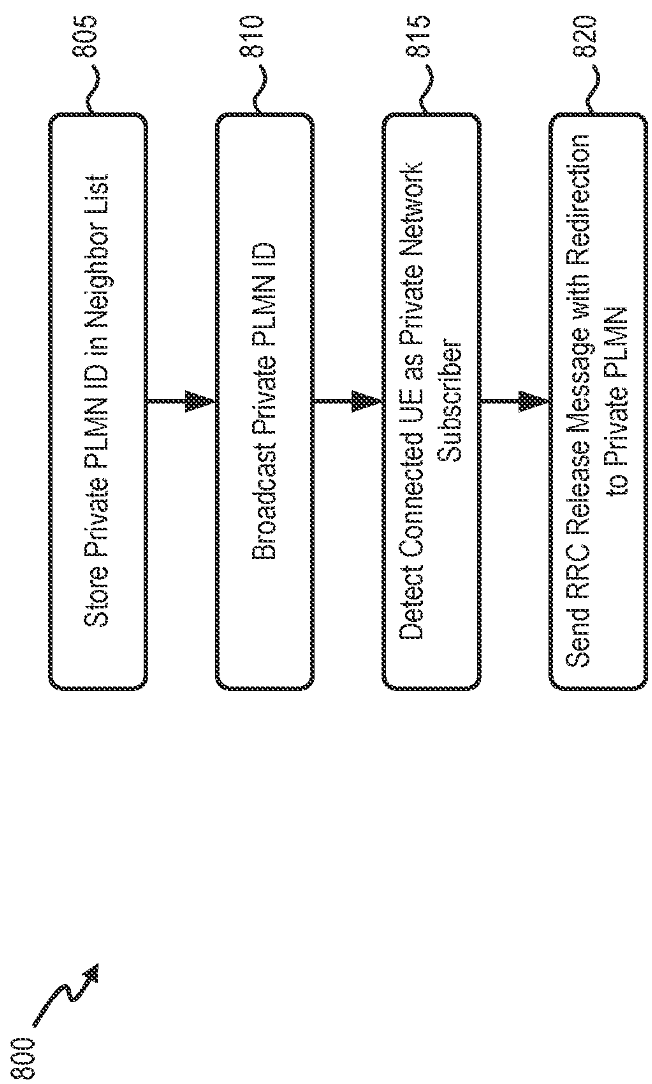

800
805
Store Private PLMN ID in Neighbor List
810
Broadcast Private PLMN ID
815
Detect Connected UE as Private Network Subscriber
820
Send RRC Release Message with Redirection to Private PLMN

FIG. 8

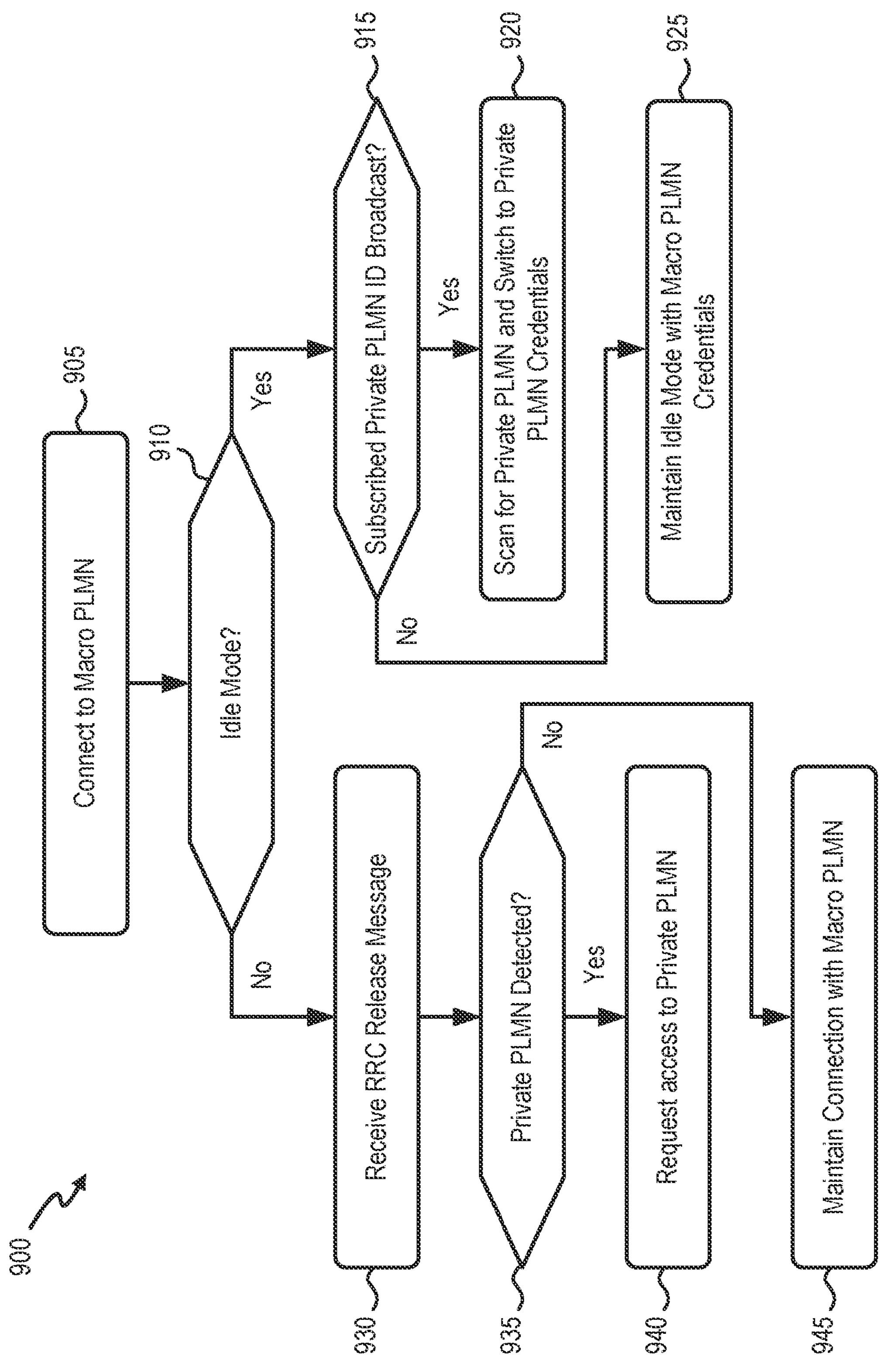

900
Connect to Macro PLMN
905
Idle Mode?
910
Yes
Subscribed Private PLMN ID Broadcast?
915
Yes
Scan for Private PLMN and Switch to Private PLMN Credentials
920
No
Maintain Idle Mode with Macro PLMN Credentials
925
No
Receive RRC Release Message
930
Private PLMN Detected?
935
Yes
Request access to Private PLMN
940
No
Maintain Connection with Macro PLMN
945

FIG. 9

# NETWORK DRIVEN PRIVATE AND MACRO NETWORK INTERWORKING

## BACKGROUND INFORMATION

Public Land Mobile Networks (PLMNs) are traditionally designed to support mobile devices over an extensive geographic area (e.g., national or regional coverage). Private mobile networks, in contrast, may use similar wireless technologies but may be constrained to a smaller geographic area. Private mobile networks may be used for a variety of reasons, such as to assure better local coverage, low latency, privacy, security, or flexibility to launch specific services. For private mobile network subscribers, there may be a need to fall back to a PLMN when private network coverage is not available and automatically return to the private network coverage when it becomes available.

## BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a diagram illustrating concepts described herein;

FIG. **2** illustrates an example network in which the systems and methods described herein may be implemented;

FIG. **3** depicts example components of a device described herein;

FIG. **4** is a diagram illustrating communications to identify private network capabilities of a UE device, according to an implementation;

FIG. **5** is a diagram illustrating communications to direct a User Equipment (UE) device from a macro network to a private network when the UE device is in an idle mode, according to an implementation;

FIG. **6** is a diagram illustrating communications to redirect a UE device from a macro network to a private network when the UE device is in a connected mode, according to an implementation;

FIG. **7**A is an example of neighboring private access station information that may be included in a System Information Block, according to an implementation;

FIG. **7**B is an example of neighboring private access station information that may be included in a Radio Resource Control (RRC) release message, according to an implementation; and

FIGS. **8** and **9** are flow diagrams illustrating processes for providing network driven private and macro network interworking, according to implementations described herein.

## DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

The following detailed description refers to the accompanying drawings. The same reference numbers in different drawings may identify the same or similar elements.

A Public Land Mobile Network (PLMN) operator may offer services to private network subscribers to allow the private network subscribers to utilize the PLMN and its network services based on a policy that is established for a respective private network (e.g., as differentiated by a PLMN identifier (ID) or another private network identifier). There are an increasing number of use cases that require tight interworking between public networks (e.g., PLMNs) and private networks on the same device.

For example, user equipment (UE) devices can be configured to be able to use both a public network (also referred to as a macro network) and a private network. The private network typically is assigned higher access priority than that assigned to the public network. That is, whenever the private network is available, the UE device should access the private network using credentials for the private network, even if the macro network is also available at the same location. Whenever the private network is not available, the UE device can receive services from the macro network using the macro network credentials.

For operators providing both macro network and private network services, the private network and macro network can be independent of each other (e.g., unique PLMN IDs, unique credentials, unique home subscriber server (HSS)/unified data management (UDM), unique backhaul network, etc.). A UE device can be provisioned with different network credentials (e.g., different IMSIs) to access the private network and the macro network, respectively. In a mobility context for private network subscribers, it is desirable to have an automatic, seamless switch between the private network and the public network with minimal delays.

One possible approach to providing a solution to service these use cases may be using a device-based approach, where a UE device performs regular scans for a highest priority network. Periodic scanning for the higher priority network (e.g., the private network in this case) can negatively impact battery life for the UE device, especially when the private network is not available in most locations. Periodic scanning also causes delays, since the trigger of higher priority network selection is based on a configurable timer, which could use scanning intervals of up to six minutes or more.

Another approach to providing a solution to service these use cases may be using a network-based approach, where the mobile network redirects the UE device from the macro network to the private network when the private network is available (whether the UE device is in either an idle mode or RRC connected mode). This network approach may entail providing network knowledge or a neighbor list in the macro network. The neighbor list may identify private networks in the coverage area of a macro network cell. In contrast with a device-based approach, the network-based approach may provide faster transitions to private networks and may be more battery friendly for the UE device.

Systems and methods described herein provide a network-based solution to efficiently transition private network subscribers from a macro network to a private network. A network device (e.g., an access node or base station) in the macro network stores a neighbor list identifying a private network that has overlapping coverage with a cell for the macro network. The network device detects that a UE device connected to the network device has a subscription to the private network and in response to the detecting, sends a release message to the UE device. The release message redirects the UE device to the private network.

FIG. **1** illustrates concepts described herein. As shown in FIG. **1**, an environment **100** may include a UE device **105** (also referred to as UE **105**), a private network coverage areas **110-1** and **110-2**, and a public network coverage area **115**.

Private network coverage areas **110-1** and **110-2** (referred to collectively as "private network coverage areas **110**") may be included within the larger public network coverage area **115**. Private network coverage areas **110-1** and **110-2** may be associated with a private network identifier (e.g., a PLMN ID, such as "PLMN-A"), and public network coverage area **115** may be associated with a public network identifier (e.g., "PLMN-B").

UE device **105** (e.g., a smartphone or another mobile cellular device) may be configured with a subscription to both the private network, indicated by PLMN-A, and the public network, indicated by PLMN-B. Assume for UE device **105** that the private network (PLMN-A) is assigned a higher priority than a priority assigned to the public network (PLMN-B). Whenever UE **105** is within coverage areas **110**, UE **105** should access the private network. Whenever UE **105** is outside of coverage areas **110**, the UE **105** can receive services from the public network. To receive services within the different coverage areas **110** and **115**, UE **105** may transition between using credentials for the private network (e.g., a SIM "A") and credentials for the public network (e.g., SIM "B").

As shown in FIG. **1**, UE device **105**, in a mobility context, may move within public network coverage area **115** into and out of private network coverage areas **110**. For example, at transition **120**, UE device **105** may transition from public network coverage area **115** to private network coverage area **110-1** while in a connected state (e.g., a Radio Resource Control (RRC) connected mode). UE device **105** would likely not detect any signal degradation for PLMN-B when UE enters private network coverage area **110-1**, since UE device **105** remains in public network coverage area **115**. Thus, according to implementations described herein, devices in PLMN-B may provide a network-based trigger, in the form of a connection release message from PLMN-B, to quickly initiate a system change in UE device **105** from PLMN-B to PLMN-A.

At transition **130**, UE device **105** may transition out of private network coverage area **110-1** and back into public network coverage area **115**. UE device **105** would detect signal degradation for PLMN-A when UE exits private network coverage area **110-1** and automatically perform a system change from PLMN-A to PLMN-B.

At transition **140**, assume UE device **105** has entered into an idle mode, due to inactivity. UE device **105** may transition from public network coverage area **115** to private network coverage area **110-2**. Again, UE device **105** would likely not detect any signal degradation for PLMN-B when UE enters private network coverage area **110-2**. Thus, network devices for PLMN-B may provide a trigger, in the form of a broadcast signal, to initiate a system change in UE device **105** from SIM "B" to SIM "A," so that UE device **105** will initiate a connection to PLMN-A upon exiting the idle mode.

FIG. **2** is simplified diagram of a network environment **200** in which systems and methods described herein may be implemented. As shown in FIG. **2**, environment **200** may include UE device **105**, private network **110**, and a macro network **115**. Private network **110** may include a private RAN **220** and a private core network **230**. Macro network **115** may include a public RAN **225** and a public core network **235**. Network environment **200** may also include a one or more data networks **250**.

UE device **105** may include any device with long-range (e.g., cellular or mobile wireless network) wireless communication functionality. For example, UE device **150** may include a handheld wireless communication device (e.g., a mobile phone, a smart phone, a tablet device, etc.); a wearable computer device (e.g., a wristwatch computer device, etc.); a portable computer; a customer premises equipment (CPE) device, such as a set-top box or a digital media player, a WiFi access point, a smart television, etc.; a mobile device; a portable gaming system; global positioning system (GPS) device; a home appliance device; a home monitoring device; a fixed wireless access (FWA) device; and/or any other type of computer device with wireless communication capabilities. Other examples of UE device **105** may include a machine-type communication (MTC) device, an Unmanned Aerial Vehicle (UAV), and an autonomous terrestrial vehicle.

UE **105** may be registered with private network **110** and include a Subscriber Identity Module (SIM), illustrated as SIM "A" in FIG. **2**. A SIM may include, for example, an embedded SIM (eSIM), a Universal Integrated Circuit Card (UICC), or embedded UICC (eUICC). The SIM may store a user (or a subscriber) credentials associated with accessing private network **110**. Furthermore, to enable UE **105** to receive services from other networks (e.g., macro network **115**) when outside the coverage area of private network **110**, UE **105** may also include a separate SIM or other credentials for accessing macro network **115**, illustrated as SIM "B" in FIG. **2**.

Private network **110** may include a private wireless network that includes a private radio access network (RAN) **220**, a private core network **230**. For example, private network **110** may be associated with an enterprise or corporate entity. RAN **220** may allow UE **105** to access core network **230**. To do so, RAN **220** may establish and maintain, with participation from UE **105**, an over-the-air channel with UE **105** and maintain backhaul channels with core network **230**. RAN **220** may convey information through these channels, from UE **105** to core network **230** and vice versa.

RAN **220** may include an LTE radio network, a Next Generation (NG) radio access network (e.g., Fifth Generation (5G) radio access network) and/or another advanced radio network. These radio networks may operate in many different frequency ranges, including millimeter wave (mmWave) frequencies, sub 6 GHz frequencies, and/or other frequencies. RAN **220** may include multiple wireless stations (e.g., eNodeBs, gNodeBs, or other base stations), Central Units (CUs), Distributed Units (DUs), Radio Units (RUs), and/or Integrated Access and Backhaul (IAB) nodes, referred to collectively herein as access stations **210**. Access station **210** may establish and maintain over-the-air channels with UEs **105** and backhaul channels with core network **230**.

Core network **230** may include a local area network (LAN), a wide area network (WAN), a metropolitan area network (MAN), an optical network, a cable television network, a satellite network, a wireless network (e.g., a Code Division Multiple Access (CDMA) network, a general packet radio service (GPRS) network, an LTE network (e.g., a 4G network), a 5G network, an ad hoc network, an intranet, a PLMN, or a combination of networks. Core network **230** may allow the delivery of Internet Protocol (IP) services to UE **105** and may interface with other networks, such as data network **250**.

Core network **230** may include various core devices **240**. Depending on the implementation, core devices **240** may include 5G core network components (Unified Data Management (UDM) function, an Access and Mobility Management Function (AMF), an Authentication Server Function (AUSF), and a Unified Data Repository (UDR), etc.) and/or 4G core network components (e.g., a Serving Gateway (SGW), a Packet data network Gateway (PGW), a Mobility Management Entity (MME), a Home Subscriber Server (HSS), an Authentication Authorization and Accounting (AAA) server, a Policy and Charging Rules Function (PCRF), etc.). Core devices **240** may also include other 4G and/or 5G core network components, including combined 4G/5G network components.

Data network **250-1** may include one or more networks that are external to core network **230**. In some implementations, data network **250-1** may include packet data networks, such as an Internet Protocol (IP) network. In another implementation, data network **250-1** may be part of core network **230**. Data network **250-1** may provide particular network services, such as, for example, Voice-over-IP (VoIP) services, messaging services, video services, etc.

Macro network **115** (also referred to herein as "public network **115**") may include a PLMN associated with a mobile network operator (MNO). Macro network **115** may include a public RAN **225** and a public core network **235**. Public RAN **225** and a public core network **235** may include components similar to those described above in connection with private RAN **220** and a private core network **230**. For example, public RAN **225** may include one or more access stations **215** that support an LTE radio network, a NG RAN, and/or another advanced radio network. Public core network **235** may include core devices **245**. Similar to core devices **240** described above, core devices **245** may include core network components, including 4G or 5G core network components, future generation components and/or combined network components. Macro network **115** may include additional components that are not shown in FIG. **2** for simplicity.

Generally, macro network **115** may have a larger coverage area (e.g., regional, national, international) than private network **110**. For example, access stations **215** of macro network **115** may provide numerous overlapping cells (or coverage areas) and serve as a fallback network for private network **110**. Thus, while private RAN **220** and public RAN **225** are shown separately in FIG. **2**, in many geographic areas, coverage of public RAN **225** will overlap coverage of private RAN **220**.

FIG. **3** is a diagram illustrating exemplary components of a device **300**. Device **300** may correspond to or include, for example, UE device **105**, access stations **210/215**, core devices **240/245**, or another component of network environments **100/200**. Alternatively or additionally, UE device **105**, access station **210/215**, core device **240/245**, or the others components of network environment **100** may include one or more devices **300** and/or one or more components of device **300**.

Device **300** may include a bus **310**, a processor **320**, a memory **330**, an input component **340**, an output component **350**, and a communication interface **360**. Although FIG. **3** shows exemplary components of device **300**, in other implementations, device **300** may contain fewer components, additional components, different components, or differently arranged components than those depicted in FIG. **3**. For example, device **300** may include one or more switch fabrics instead of, or in addition to, bus **310**. Additionally, or alternatively, one or more components of device **300** may perform one or more tasks described as being performed by one or more other components of device **300**.

Bus **310** may include a path that permits communication among the components of device **300**. Processor **320** may include a processor, a microprocessor, or processing logic that may interpret and execute instructions. Memory **330** may include any type of dynamic storage device that may store information and instructions, for execution by processor **320**, and/or any type of non-volatile storage device that may store information for use by processor **320**. Input component **340** may include a mechanism that permits a user to input information to device **300**, such as a keyboard, a keypad, a button, a switch, etc. Output component **350** may include a mechanism that outputs information to the user, such as a display, a speaker, one or more light emitting diodes (LEDs), etc.

Communication interface **360** may include a transceiver that enables device **300** to communicate with other devices and/or systems via wireless communications, wired communications, or a combination of wireless and wired communications. For example, communication interface **360** may include mechanisms for communicating with another device or system via a network. Communication interface **360** may include an antenna assembly for transmission and/or reception of RF signals. For example, communication interface **360** may include one or more antennas to transmit and/or receive RF signals over the air. Communication interface **360** may, for example, receive RF signals and transmit them over the air to UE device **105**/RAN **220**, and receive RF signals over the air from RAN **220**/UE device **105**. In one implementation, for example, communication interface **360** may communicate with a network and/or devices connected to a network. Alternatively or additionally, communication interface **360** may be a logical component that includes input and output ports, input and output systems, and/or other input and output components that facilitate the transmission of data to other devices.

Device **300** may perform certain operations in response to processor **320** executing software instructions contained in a computer-readable medium, such as memory **330**. A computer-readable medium may be defined as a non-transitory memory device. A memory device may include space within a single physical memory device or spread across multiple physical memory devices. The software instructions may be read into memory **330** from another computer-readable medium or from another device. The software instructions contained in memory **330** may cause processor **320** to perform processes described herein. Alternatively, hardwired circuitry may be used in place of or in combination with software instructions to implement processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

FIG. **4** is a diagram illustrating communications to identify private network capabilities of a UE device. More particularly, FIG. **4** illustrates communications in a portion **400** of network environment **200** that includes access station **215** of public RAN **225** and UE device **105**.

According to an embodiment, access station **215** may store a "knowledge/neighbor" list **405** of private networks that are co-located in the coverage area (or cell) of access station **215**. Neighbor list **405** may be configured as part of a provisioning process when a private network is set up, for example, by a mobile network operator. The neighbor list **405** may identify PLMN IDs, priority levels, RAT-types, and/or other information for each private network that is available within the coverage area of access station **215**. As described further herein, the neighbor list **405** may be used to match connected UE devices **105** with higher priority private networks in the coverage area of access station **215**. Access station **215** may also include logic that categorizes or determines facets associated with the attachment to UE device **105**, such as the type of UE device **105** (e.g., an NB-IoT device versus a user device, etc.), the mobility characteristics of UE device **105** (e.g., stationary device, high mobility, low mobility, etc.), the access mode used by UE device **105** to attach to access station **215** (e.g., LTE, eMTC mode A, eMTC mode B, NB-IoT, etc.), and/or end device capability information pertaining to UE device **105**.

According to an implementation, access station **215** may obtain end device capability information from UE device **105**. For example, during an attachment procedure, access station **215** may transmit a control/signaling message, such as a UE capability enquiry message **410**, to UE device **105**. Typically, UE capability enquiry message **410** is a request for a UE device to list its capabilities regarding radio access technologies (RATs) (e.g., E-UTRA, UTRA, CDMA 2000, GERAN, etc.). In implementations described herein, UE capability enquiry message **410** may be configured to request a list of PLMNs (e.g., private and public network IDs) supported by a UE device. For example, a standard format of UE capability enquiry message **410** may be modified to include an additional field to request a supported PLMN list (e.g., requestSupportedPLMNlist).

In response to receiving UE capability enquiry message **410**, UE device **105** may generate and transmit a UE capability information message **420** including UE-Network-Capability, to access station **215**. The UE-Network-Capability information may be formatted to respond to the requestSupportedPLMNlist of UE capability enquiry message **410**. The UE-Network-Capability information may include a list of PLMN IDs to which UE device **105** is subscribed. In the example of UE device **105** of FIG. **2**, the UE-Network-Capability information may include a PLMN ID for private network **110**.

FIG. **5** is a diagram illustrating communications to direct a UE device from a macro network to a private network when the UE device is in an idle mode. More particularly, FIG. **5** illustrates communications in a portion **500** of network environment **200** that includes access station **215** of public RAN **225**, access station **210** of private RAN **220**, and UE device **105**.

For UE devices capable of operating with both macro network **115** and private network **110**, scanning for a private RAN by the UE device **105** located in a limited private network deployment environment may be inefficient. However, without sufficient information to identify available neighboring private RAN **220** cells and/or reselection priorities that are based on network design and loading, UE device **105** may encounter protracted mode transition (e.g., idle mode to connected mode) and/or system acquisition times, potentially negatively impacting the user experience. Accordingly, in areas having neighboring and/or overlapping private RAN and public RAN cells, access station **215** may transmit information to expedite UE device **105** establishing wireless connections with a neighboring private RAN **220** for private network **110**.

As shown in FIG. **5**, access station **215** may broadcast **510** a private PLMN ID as a neighbor of access station **215**. The private PLMN ID may be obtained, for example, from the stored neighbor list **405** described above in connection with FIG. **4**. For example, access station **215** may broadcast over a system information block (SIB) information regarding one or more private network access stations. In one implementation, access station **215** may broadcast a PLMN ID for neighboring access station **210** (e.g., PLMN ID of the private network to which the neighboring access station **210** belongs). The broadcast information may be provided over a modified Information Element (IE) SIB Type **24** (hereinafter "SIB **24**"). In another implementation, access station **215** may broadcast a PLMN ID for a neighboring private access station **210** over a new or different SIB (e.g., a "SIB **25**").

The SIB **25** or modified SIB **24** (or another SIB) may contain information relevant for private network cell selection/reselection. SIB **24** may indicate "network neighbors" using the respective private network PLMN IDs as the network indicator. FIG. **7**A provides an illustration of example information **710** that may be included for a new SIB **25**. Similar information may be included in a modified SIB **24** or another SIB.

In the example, of FIG. **5**, broadcast **510** from access station **215** may include the PLMN ID of private network **110** as a network neighbor. The private PLMN-ID in the SIB of broadcast **510** may be used by the UE device **105** receiving the broadcast information to expedite idle mode transitions from macro network **115** to private network **110** and ensure that UE device **105** selects/reselects to the higher priority private network based on subscriber priorities, for example.

Thus, as indicated at referenced **520**, UE device **105** may detect that private PLMN ID is broadcasted (e.g., in a SIB) as a neighbor. UE device **105** may provide the received neighbor list to its UICC and/or local profile agent (LPA) to scan **530** for private network coverage from access station **210**, since access station **215** and access station **210** may not have the same coverage. Assuming UE device **105** scans and finds private RAN **220**, the UICC/LPA of UE device **105** may switch **540** to using private network credentials (e.g., SIM "A") for exiting RRC idle mode.

FIG. **6** is a diagram illustrating communications to redirect a UE device from a macro network to a private network when the UE device is in a connected mode. FIG. **6** illustrates communications in portion **500** of network environment **200**.

In FIG. **6**, assume that UE device **105** is already in RRC connected mode with access station **215** and that UE device **150** is a private-network-capable device (e.g., as determined through separate provisioning or the UE capability identification process of FIG. **4**). Through the provisioning and setup of the private PLMN **110**, private RAN **220** would be included in a neighboring PLMN list (e.g., neighbor list **405**) for public RAN **225**. Public RAN **225** would be aware there is a co-located private network neighbor (e.g., private RAN **220**) for UE device **105**. For example, access station **215** may compare UE capability information message **420** (FIG. **4**) with stored PLMN neighbor list **405** for access station **215** and detect **600** a co-located private network neighbor for UE device **105**.

Access station **215** may send a RRC release message **610** to UE device **105**, to redirect UE device **105** to the private network. According to an implementation, a RRC release message **610** may include an additional field added to a conventional RRC release message to identify another PLMN as a redirection target if the UE device supports the high priority PLMN neighbor. Thus, an RRCRelease message may include redirectedNetworkInfo indicating redirection to a private PLMN (with the same or different radio access technology using existing redirectionCarrierInfo). The release message may include a PLMN-ID or another type of identifier for the private network. FIG. **7**B provides an illustration of example information **720** that may be included in a modified RRCRelease message.

The RRC release procedure can be measurement based or non-measurement based (herein referred to as blind procedure). For a measurement based release procedure, access station **215** may send RRC release message **610** with a measurement trigger to ask UE device **105** to measure signal strength from private RAN **220** before conducting redirection. In response, UE device **105** may measure **620** signal strength for the private PLMN. For blind redirection (e.g., in confirmed co-located cases), RAN **225** may instruct a blind redirection (e.g., that doesn't require UE device **105** to first measure signal strength of private RAN **220**).

When UE device **105** receives the RRC release message **610** with redirection target of the private PLMN and UE device **105** has credentials (e.g., SIM "A") to access the private PLMN, UE device **105** will attempt to access private RAN **220** (e.g., via providing an RRC connection request **630**).

FIG. **8** is a flow diagram illustrating a process **800** for providing network driven private and macro network interworking. In one implementation, process **800** may be implemented by devices in a macro PLMN, such as macro network **115**. In another implementation, process **800** may be implemented by devices in macro network **115** in conjunction with one or more other devices in network environment **100**, such as UE device **110**. Process **800** may address network-driven private and macro network interworking for UE devices in an idle mode or connected mode.

Process **800** may include storing a private PLMN ID in a neighbor list (block **805**) and broadcasting the private PLMN ID (block **810**). For example, access station **215** of public RAN **225** may store a "knowledge/neighbor" list of private PLMNs that are co-located in the coverage area (or cell) of access station **215**. Additionally, access station **215** may broadcast **510** a private PLMN ID for a neighbor of access station **215**. The private PLMN ID may be obtained, for example, from the stored neighbor list and broadcast over a new or modified system information block (SIB). The private PLMN ID may be used by a UD device **105** in idle mode to initiate a switch to credentials for the private PLMN.

Process **800** may further include detecting a connected UE as a private network subscriber of a neighboring private PLMN (block **815**) and sending an RRC release message with redirection to the private PLMN (block **820**). For example, access station **215** may provide a UE capability enquiry message **410** that includes a request for a supported PLMN list. In response to the UE capability enquiry message **410**, a UE device **105** may generate and transmit to access station **215** a UE capability information message **420** including a list of PLMN IDs to which UE device **105** is subscribed. Based on stored neighbor list **405** and UE capability information message **420**, access station **215** may compare UE capability information with neighbor PLMN-IDs and detect a co-located private network neighbor for UE device **105**. Assuming access station **215** identifies a private PLMN with the matching PLMN ID, access station **215** may send a modified RRC release message **610** with the PLMN ID for the private network as redirection target for UE device **105**.

FIG. **9** is a flow diagram illustrating a process **900** for providing network driven private and macro network interworking. In one implementation, process **900** may be implemented by UE devices (e.g., UE device **105**) that are configured with subscriptions for both private networks and public networks. In another implementation, process **900** may be implemented by UE device **105** in conjunction with one or more other devices in network environment **100**, such as devices in macro network **115**. Process **900** may address UE actions to support network-driven private and macro network interworking.

Process **900** may include connecting to a macro PLMN (block **905**) and identifying if the UE device enters an idle mode (block **910**). For example, UE device **105** may attach to public RAN **225** and connect to macro network **115**. Depending on an activity level, UE device **105** may enter an idle mode.

If the UE device enters an idle mode (block **910**—Yes), it may be determined if a subscribed private PLMN ID is being broadcast as an ID of a neighbor PLMN (block **915**). For example, access station **215** may broadcast **510** (e.g., in a new or modified SIB) a private PLMN ID for a neighbor of access station **215**.

If a subscribed private PLMN ID is being broadcast as the ID of a neighbor PLMN (block **915**—Yes), the UE device may scan for the private PLMN and switch to private PLMN credentials (block **920**). For example, UE device **105** may recognize the private PLMN ID broadcast by access station **215**. A UICC/LPA of UE device **105** scan for private PLMN coverage. Assuming UE device **105** scans and finds private RAN **220**, the UICC/LPA of UE device **105** may switch to using its private PLMN credentials.

If a subscribed private PLMN ID is not being broadcast as a neighbor PLMN (block **915**—No), the UE device may maintain the idle mode with the macro PLMN credentials (block **925**). For example, UE device **105** would not switch credentials from the public RAN **225** if UE device did not recognize a private PLMN ID (e.g., as a subscriber) or does not detect sufficient signal strength from private RAN **220**.

If the UE device does not enter an idle mode (block **910**—No), UE device may receive a RRC release message (block **930**) and determine if the private PLMN can be detected (block **935**). For example, access station **215** may compare UE capability information message **420** with a stored PLMN neighbor list **405** for access station **215** and detect a co-located private network neighbor for UE device **105**. Based on this detection, access station **215** may send a RRC release message that identifies the private PLMN as a redirection target. UE device **105** may receive the RRC release message and may measure a signal strength for the private PLMN.

If the private PLMN can be detected (block **935**—Yes), the UE device may request access to the private PLMN (block **940**). For example, when UE device **105** receives the RRC release message with a redirection target of the private PLMN and UE device **105** has credentials to access the private PLMN, UE device **105** will attempt to access private RAN **220**.

If the private PLMN cannot be detected (block **935**—No), the UE device may maintain a connection with the macro PLMN (block **945**). For example, UE device **105** would not switch from the public RAN **225** if UE device did not detect sufficient signal strength from private RAN **220** (or if an attempted connection to the private PLMN fails).

Systems and methods described herein provide a network-based solution to efficiently transition private network subscribers from a macro network to a private network. In contrast with a device-based approach, this network-based approach provides fast transitions to private networks and is more battery friendly for the UE device.

The foregoing description of embodiments provides illustrations but is not intended to be exhaustive or to limit the embodiments to the precise form disclosed. In the preceding description, various embodiments have been described with reference to the accompanying drawings. However, various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the broader scope of the invention as set forth in the claims that follow. The description and drawings are accordingly to be regarded as illustrative rather than restrictive.

In addition, while series of blocks and signals have been described with regard to the processes illustrated in FIGS. **4-6**, **8** and **9**, the order of the blocks and/or signals may be modified according to other embodiments. Further, non-dependent blocks may be performed in parallel. Additionally, other processes described in this description may be modified and/or non-dependent operations may be performed in parallel.

The embodiments described herein may be implemented in many different forms of software executed by hardware. For example, a process or a function may be implemented as "logic" or as a "component." The logic or the component may include, for example, hardware (e.g., processor **320**, etc.), or a combination of hardware and software. The embodiments have been described without reference to the specific software code since the software code can be designed to implement the embodiments based on the description herein and commercially available software design environments/languages.

As set forth in this description and illustrated by the drawings, reference is made to "an exemplary embodiment," "an embodiment," "embodiments," etc., which may include a particular feature, structure or characteristic in connection with an embodiment(s). However, the use of the phrase or term "an embodiment," "embodiments," etc., in various places in the specification does not necessarily refer to all embodiments described, nor does it necessarily refer to the same embodiment, nor are separate or alternative embodiments necessarily mutually exclusive of other embodiment(s). The same applies to the term "implementation," "implementations," etc.

The terms "a," "an," and "the" are intended to be interpreted to include one or more items. Further, the phrase "based on" is intended to be interpreted as "based, at least in part, on," unless explicitly stated otherwise. The term "and/or" is intended to be interpreted to include any and all combinations of one or more of the associated items. The word "exemplary" is used herein to mean "serving as an example." Any embodiment or implementation described as "exemplary" is not necessarily to be construed as preferred or advantageous over other embodiments or implementations.

Use of ordinal terms such as "first," "second," "third," etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another, the temporal order in which acts of a method are performed, the temporal order in which instructions executed by a device are performed, etc., but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish the claim elements.

Additionally, embodiments described herein may be implemented as a non-transitory storage medium that stores data and/or information, such as instructions, program code, data structures, program modules, an application, etc. The program code, instructions, application, etc., is readable and executable by a processor (e.g., processor **320**) of a computational device. A non-transitory storage medium includes one or more of the storage mediums described in relation to memory **330**.

To the extent the aforementioned embodiments collect, store or employ personal information provided by individuals, it should be understood that such information shall be used in accordance with all applicable laws concerning protection of personal information. Additionally, the collection, storage and use of such information may be subject to consent of the individual to such activity, for example, through well known "opt-in" or "opt-out" processes as may be appropriate for the situation and type of information. Storage and use of personal information may be in an appropriately secure manner reflective of the type of information, for example, through various encryption and anonymization techniques for particularly sensitive information.

No element, act, or instruction described in the present application should be construed as critical or essential to the embodiments described herein unless explicitly described as such.

What is claimed is:

**1**. A method, comprising:

storing, at an access station for a macro Public Land Mobile Network (PLMN), a neighbor list identifying a private network that has overlapping coverage with a cell for the access station;

detecting, by the access station, that a user equipment (UE) device connected to the access station has a subscription to the private network, wherein the detecting includes:

sending, during an attachment procedure, a UE capability inquiry to the UE device,

receiving a UE network capability response from the UE device, and

comparing the UE network capability response with the neighbor list; and

sending, by the access station and in response to the detecting, a release message to the UE device, wherein the release message includes a PLMN identifier (ID) as a redirection target to the private network.

**2**. The method of claim **1**, further comprising:

broadcasting, by the access station, an identifier for the private network.

**3**. The method of claim **2**, wherein broadcasting the identifier includes:

broadcasting a system information block that includes a PLMN-ID for the private network.

**4**. The method of claim **2**, wherein broadcasting the identifier for the private network triggers the UE device in an idle mode to:

scan for signals from the private network, or

switch to using credentials for the private network.

**5**. The method of claim **1**, wherein sending the release message further comprises:

sending, to the UE device, a measurement trigger requesting the UE device to measure signal strength from the private network before conducting a redirection.

**6**. The method of claim **1**, wherein the UE capability inquiry is configured to request a list of PLMNs supported by the UE device.

**7**. The method of claim **1**, wherein the UE network capability response includes a list of PLMN IDs that identify PLMNs to which the UE device is subscribed.

**8**. The method of claim **1**, wherein the subscription establishes a priority associated with the private network to be higher than a priority associated with the macro PLMN.

**9**. The method of claim **1**, wherein the macro PLMN and the private network require different credentials from the UE device.

**10**. A network device comprising:

a processor configured to:

store, in a macro Public Land Mobile Network (PLMN), a neighbor list identifying a private network that has overlapping coverage with a cell for the macro PLMN;

detect that a user equipment (UE) device connected to the network device has a subscription to the private network, wherein the detecting further includes the processor configured to:

send, during an attachment procedure, a UE capability inquiry to the UE device,

receive a UE network capability response from the UE device, and

compare the UE network capability response with the neighbor list; and

send, in response to the detecting, a release message to the UE device, wherein the release message includes a PLMN identifier (ID) as a redirection target to the private network.

**11**. The device of claim **10**, wherein the processor is further configured to:

broadcast an identifier for the private network.

**12**. The device of claim **10**, wherein the network device includes an access station for a radio access network (RAN).

**13**. The device of claim **10**, wherein when sending the release message, the processor is further configured to:

send, to the UE device, a measurement trigger requesting the UE device to measure signal strength from the private network before conducting a redirection.

**14**. The device of claim **10**, wherein the UE network capability response includes a list of PLMN IDs that identify PLMNs to which the UE device is subscribed.

**15**. The device of claim **10**, wherein the macro PLMN uses different credentials from the UE device than the private network.

**16**. The device of claim **10**, wherein the release message includes a Radio Resource Control (RRC) release message.

**17**. A non-transitory computer-readable medium comprising processor-executable instructions that, wherein when executed by a processor, cause the processor to:

store, at an access station for a macro Public Land Mobile Network (PLMN), a neighbor list identifying a private network that has overlapping coverage with a cell for the access station;

detect that a user equipment (UE) device connected to the access station has a subscription to the private network, wherein the detecting further causes the processor to:

send, during an attachment procedure, a UE capability inquiry to the UE device,

receive a UE network capability response from the UE device, and

compare the UE network capability response with the neighbor list; and

send, in response to the detecting, a release message to the UE device, wherein the release message includes a PLMN identifier (ID) as a redirection target to the private network.

**18**. The non-transitory computer-readable medium of claim **17**, further comprising instruction that cause the processor to:

broadcast an identifier for the private network in a system information block (SIB).

**19**. The non-transitory computer-readable medium of claim **17**, wherein the instructions to detect that the UE device has the subscription to the private network further comprise instruction that cause the processor to:

receive, from the UE device, a list of PLMNs to which the UE device is subscribed.

**20**. The non-transitory computer-readable medium of claim **17**, wherein the release message includes a Radio Resource Control (RRC) release message.

* * * * *